SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	S Sandhiya	**Reg. No.:**	192321015
Section / Batch:	D	**Date:**	5-10-26

Code of Conduct

I S Sandhiya(!92321015) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S Sandhiya

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.

- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Annexure A – Analytical Problem Solving

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

(a) PEAS Description [4 Marks]

PEAS	Description
P – Performance Measure	+1 for every clean square at the end of each time step; penalty of p for every Left/Right move.
E – Environment	Two squares A and B; initially both are Dirty. Dirt never reappears.
A – Actuators	Suck, Left, Right, NoOp
S – Sensors	Detects current location (A/B) and whether the current square is Dirty/Clean.

(b) Agent X – Simple Reflex [5 Marks]

Initial condition

- Position = A
- A = Dirty
- B = Dirty
- p = 1

Agent X rule:

- Dirty → Suck
- Clean → Move to the other square

Time	Position	Percept	Action	Clean squares at end	Move penalty	Step score
1	A	Dirty	Suck	1	0	1
2	A	Clean	Right	1	1	0
3	B	Dirty	Suck	2	0	2
4	B	Clean	Left	2	1	1
5	A	Clean	Right	2	1	1
6	B	Clean	Left	2	1	1

Total score of X

$$1 + 0 + 2 + 1 + 1 + 1 = \boxed{6}$$

So,

$$\boxed{\text{Score(X)} = 6}$$

(c) Agent Y – Reflex with Internal State [4 Marks]

Agent Y rule:

- Dirty → Suck
- Clean + both not cleaned → Move
- Clean + remembers both cleaned → NoOp

Time	Position	Percept	Action	Clean squares at end	Move penalty	Step score
1	A	Dirty	Suck	1	0	1
2	A	Clean	Right	1	1	0
3	B	Dirty	Suck	2	0	2
4	B	Clean	NoOp	2	0	2
5	B	Clean	NoOp	2	0	2
6	B	Clean	NoOp	2	0	2

Total score of Y

$$1 + 0 + 2 + 2 + 2 + 2$$
$$\boxed{\text{Score(Y)} = 9}$$

So Agent Y scores 9, while Agent X scores 6.

(d) Score as a Function of p[3 Marks]

Agent X

Agent X makes 4 moves:

- Time 2 → Right
- Time 4 → Left
- Time 5 → Right
- Time 6 → Left

Total clean-square contribution:

$$1 + 1 + 2 + 2 + 2 + 2 = 10?$$

Careful: the actual clean-square counts are:

$$1 + 1 + 2 + 2 + 2 + 2 = 10$$

Therefore:

$$\boxed{X = 10 - 4p}$$

For $p = 1$:

$$10 - 4 = \boxed{6}$$

Agent Y

Agent Y makes only 1 move.

Clean-square contribution:

$$1 + 1 + 2 + 2 + 2 + 2 = 10$$

Therefore:

$$\boxed{Y = 10 - p}$$

For $p = 1$:

$$10 - 1 = \boxed{9}$$

When is Y strictly better?

$$10 - p > 10 - 4p$$
$$-p > -4p$$
$$3p > 0$$

Therefore:

$$\boxed{p > 0}$$

(e) Which Agent is Rational? [4 Marks]

Agent Y is rational under the given performance measure.

A rational agent chooses the action that maximizes its expected performance based on its percept history and available information.

Agent Y remembers that both squares have already been cleaned. Therefore, after cleaning both squares, it chooses NoOp instead of repeatedly moving between A and B.

Agent X does not have internal memory. It only considers the current percept. Whenever it sees a clean square, its rule says to move to the other square. Therefore, even after both squares are clean, X continues moving unnecessarily and receives movement penalties.

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	**Easier value (0)**	**Harder value (1)**
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

(a) Classify each environment on all six properties [9 Marks]

Property	E1: Autonomous Taxi	E2: Part-Picking Robot	E3: Chess Agent
Observability	Partially observable – sensors cannot see everything in traffic	Partially observable – camera may miss hidden/unclear defects	Fully observable – complete chess board is visible

Number of agents	Multi-agent – other vehicles, pedestrians, etc.	Single-agent – robot performs the task independently	Multi-agent – two competing players
Determinism	Stochastic – traffic and human behavior are unpredictable	Stochastic – belt motion and detection can vary	Deterministic – legal move gives a predictable board state
Episodic / Sequential	Sequential – current driving affects future situations	Sequential – actions affect subsequent picking opportunities	Sequential – every move affects future moves
Static / Dynamic	Dynamic – traffic and surroundings change continuously	Dynamic – conveyor belt keeps moving	Semi-dynamic – board does not change by itself, but clock keeps running
Discrete / Continuous	Continuous – speed, position and steering vary continuously	Continuous – belt movement and robot motion are continuous	Discrete – finite board positions and legal moves

Easy way to remember:

- **Taxi:** Partially observable + Multi-agent + Stochastic + Sequential + Dynamic + Continuous
- **Robot:** Partially observable + Single-agent + Stochastic + Sequential + Dynamic + Continuous
- **Chess:** Fully observable + Multi-agent + Deterministic + Sequential + Semi-dynamic + Discrete

(b) Difficulty Score [4 Marks]

According to the question:

- Harder property = **1**
- Easier property = **0**
- Semi-dynamic = **0.5**

E1 – Autonomous Taxi

Property	**Classification**	**Score**
Agents	Multi-agent	
Determinism	Stochastic	1
Episodic/Sequential	Sequential	1
Static/Dynamic	Dynamic	1
Discrete/Continuous	Continuous	1
Total		**6**

$\boxed{E1 = 6}$

E2 – Part-Picking Robot

Property	**Classification**	**Score**
Observability	Partially observable	1
Agents	Single-agent	0
Determinism	Stochastic	1
Episodic/Sequential	Sequential	1
Static/Dynamic	Dynamic	1
Discrete/Continuous	Continuous	1
Total		**5**

$$\boxed{E2 = 5}$$

E3 – Chess

Property	**Classification**	**Score**
Observability	Fully observable	0
Agents	Multi-agent	1
Determinism	Deterministic	0
Episodic/Sequential	Sequential	1
Static/Dynamic	Semi-dynamic	0.5
Discrete/Continuous	Discrete	0
Total		**2.5**

$$\boxed{E3 = 2.5}$$

Ranking

$$\boxed{E1 > E2 > E3}$$

(c) Suitable Agent Architecture [4 Marks]

Hardest: E1 – Autonomous Taxi

Best choice: Learning Agent

A learning agent is most suitable because the taxi operates in a partially observable, stochastic, multi-agent, dynamic and continuous environment.

It must learn from traffic patterns, road conditions, pedestrian behavior and previous experiences.

E1 → Learning Agent

Easiest: E3 – Chess

Best choice: Goal-Based Agent

Chess has a fully observable, deterministic and discrete environment. The agent can evaluate possible moves and select moves that help achieve the goal of checkmating the opponent.

E3 → Goal-Based Agent

(d) Interpretation [3 Marks]

The complexity of an AI agent depends not only on what task it performs, but also on the properties of its environment.

For example, an autonomous taxi needs a more complex agent because it must handle:

- Partial information
- Other agents
- Uncertainty
- Continuous changes
- Sequential decisions
- Dynamic surroundings

In contrast, chess is easier to model because the board is fully observable, deterministic and discrete.

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

(a) Correct-identification rate and fooling rate [6 Marks]

Formula

Correct-identification rate:

$$\text{Correct Identification Rate} = \frac{\text{Correctly Identified}}{\text{Total Interrogations}} \times 100$$

Fooling rate:

$$\text{Fooling Rate} = 100 - \text{Correct Identification Rate}$$

Because if the judge does **not** correctly identify the machine, the machine has fooled the judge into thinking it is human.

Calculations

System	Interrogations	Correctly Identified	Correct-identification rate	Fooling rate
ELIZA	50	47	$47/50 \times 100 =$**94%**	**6%**
Chatbot B	50	36	$36/50 \times 100 =$**72%**	**28%**
System C	50	29	$29/50 \times 100 =$**58%**	**42%**

(b) Turing's 70% Criterion [5 Marks]

Turing's criterion says that the average interrogator would have no more than a 70% chance of correctly identifying the machine.

Therefore, a system satisfies the criterion if:

$$\boxed{\text{Correct-identification rate} \leq 70\%}$$

Distance from 50% guessing level

Pure guessing gives:

$$50\%$$

Calculate:

ELIZA

$$94 - 50 = \boxed{44 \text{ percentage points}}$$

Chatbot B

$$72 - 50 = \boxed{22 \text{ percentage points}}$$

System C

$$58 - 50 = \boxed{8 \text{ percentage points}}$$

System	Identification Rate	Distance above 50%
ELIZA	94%	44 pp
Chatbot B	72%	22 pp
System C	58%	8 pp

So System C is closest to the guessing level, meaning judges had the most difficulty distinguishing it from a human.

(c) Improvement in Fooling Rate [5 Marks]

ELIZA → System C

ELIZA fooling rate:

$$6\%$$

System C fooling rate:

$$42\%$$

Improvement in percentage points

$$42 - 6 = \boxed{36 \text{ percentage points}}$$

Improvement as a ratio

$$\frac{42}{6} = 7$$

Therefore, System C's fooling rate is:

$$\boxed{7\times}$$

the fooling rate of ELIZA.

So:

Fooling rate improved by 36 percentage points, or increased sevenfold.

Chatbot B → System C

Chatbot B fooling rate:

$$28\%$$

System C:

$$42\%$$

Therefore:

$$42 - 28 = \boxed{14 \text{ percentage points}}$$

So the fooling rate improved by:

$$\boxed{14 \text{ percentage points}}$$

(d) Interpretation: Acting Humanly vs Thinking Rationally [4 Marks]

The Turing Test measures whether a machine can act humanly, because the judge evaluates the machine only through its conversational behavior.

System C has a 42% fooling rate, so it can imitate human conversational behavior better than the other systems in this experiment.

However, a high fooling rate does not prove that the system is thinking, understanding, or acting rationally.

Chinese Room argument

John Searle's Chinese Room argument suggests that a system can produce appropriate language responses by following rules without necessarily understanding the meaning of the language.

Similarly, System C may produce convincing human-like responses without possessing human-like understanding or consciousness.

Therefore:

$$\boxed{\text{Acting humanly} \neq \text{necessarily thinking humanly}}$$

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

U = 100 − T − 20·L

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

Question 4: Rational Route Choice under Uncertainty

Given:

$$U = 100 - T - 20L$$

where:

- T= travel time
- $L = 1$if arrival is **after 55 minutes**
- $L = 0$otherwise

(a) Utility of Each Route in Each Outcome [4 Marks]

R1: Highway

Normal: $T = 30$, not late, so $L = 0$

$$U = 100 - 30 - 20(0) = \boxed{70}$$

Disrupted: $T = 70$, which is late, so $L = 1$

$$U = 100 - 70 - 20(1)$$
$$= 100 - 70 - 20 = \boxed{10}$$

R2: City Road

Normal: $T = 45$, not late.

$$U = 100 - 45 = \boxed{55}$$

Disrupted: $T = 60$, **late**.

$$U = 100 - 60 - 20 = \boxed{20}$$

R3: Ring Road

R3 is never disrupted and takes 50 minutes.

$$U = 100 - 50 = \boxed{50}$$

Table

Route	Outcome	Time	Late?	Utility
R1 Highway	Normal	30	No	**70**
R1 Highway	Disrupted	70	**Yes**	**10**
R2 City road	Normal	45	No	**55**
R2 City road	Disrupted	60	**Yes**	**20**
R3 Ring road	Normal	50	No	**50**

(b) Expected Utility and Rational Choice [5 Marks]

The formula for expected utility is:

$$EU = \sum P\,(\text{outcome}) \times U(\text{outcome})$$

R1: Highway

Probability of normal:

$$1 - 0.30 = 0.70$$

Therefore:

$$EU(R1) = (0.70)(70) + (0.30)(10)$$
$$= 49 + 3$$
$$\boxed{EU(R1) = 52}$$

R2: City Road

Probability of normal:

$$1 - 0.10 = 0.90$$

Therefore:

$$EU(R2) = (0.90)(55) + (0.10)(20)$$
$$= 49.5 + 2$$
$$\boxed{EU(R2) = 51.5}$$

R3: Ring Road

R3 is always 50 minutes:

$$EU(R3) = 1(50)$$
$$\boxed{EU(R3) = 50}$$

Therefore, the rational agent chooses the route with the **highest expected utility**:

$$\boxed{\text{Choose R1 – Highway}}$$

(c) New Radio Alert [4 Marks]

The new percept says the probability of disruption on R1 increases to:

$$P(\text{disruption}) = 0.80$$

Therefore:

$$P(\text{normal}) = 1 - 0.80 = 0.20$$

The utilities remain:

- Normal = 70
- Disrupted = 10

So:

$$EU(R1) = (0.20)(70) + (0.80)(10)$$
$$= 14 + 8$$
$$\boxed{EU(R1) = 22}$$

The other routes remain:

$$EU(R2) = 51.5$$
$$EU(R3) = 50$$

Therefore:

$$51.5 > 50 > 22$$

So the rational decision changes to:

$$\boxed{\text{Choose R2 – City Road}}$$

Important point:

The roads themselves did not change. The agent's information about the probability of an accident changed, so its expected utilities changed.

(d) Rational vs Omniscient Agent [4 Marks]

In part (b), the rational agent chose **R1** because:

$$EU(R1) = 52$$

was the highest expected utility.

Suppose the accident actually occurs.

R1 then takes 70 minutes, so the taxi is late.

Thus:

$$U = 100 - 70 - 20$$
$$\boxed{U = 10}$$

(e) Interpretation [3 Marks]

Rationality depends on the percept sequence because the agent uses everything it has perceived so far to make its decision.

In part (b), the agent knew:

$$P(\text{R1 disruption}) = 0.30$$

So it selected R1 because:

$$EU(R1) = 52$$

was the highest.

In part (c), the radio alert provided **new information**:

$$P(\text{R1 disruption}) = 0.80$$

This reduced:

$$EU(R1): 52 \rightarrow 22$$

Therefore, the agent changed its decision to R2.